

MORNING

[Total No. of Questions: 09]

[Total No. of Pages: 02]

Uni. Roll No.

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Program/ Course: B.Tech (Batch 2018 onward)

Semester: 3rd

Name of Subject: Mathematics-III

Subject Code: BSCS-101

Paper ID: 16013

Scientific Calculator is Allowed

Time Allowed: 3 Hours

Max. Marks: 60

NOTE:

- 1) Parts A and B are compulsory.
- 2) Part -C has Two questions Q8 and Q9. Both are compulsory, but with internal choice.
- 3) Any missing data may be assumed appropriately.

Part -A

[Marks: 02 each]

Q1.

- a) Separate $f(z) = \cosh z$ into real and imaginary parts.
- b) Solve the system of equations : $4x - 3y = 1$, $6x + 2y = 8$.
- c) Evaluate $\int_C \frac{dz}{(z-3)^3}$ where C is the circle $|z+2| = \frac{1}{2}$.
- d) Write algorithm for second degree parabola fit of a curve.
- e) Check the correctness of the statement, " Mean of a binomial distribution is 15 and S.D is 5."
- f) Write Cauchy Riemann equations in Polar form.

Part -B

[Marks: 04 each]

Q2. Fit the curve $y = a + bx$ to the following data :

x	2	5	6	9	11
y	2	4	6	9	10

Q3. Expand $f(z) = \frac{(z-2)(z+2)}{(z+1)(z+4)}$ in the region $1 < |z| < 4$.

Q4. Solve the following equations by Gauss Elimination method :

$$x + 2y - z = 3, 3x - y + 2z = 1, 2x - 2y + 3z = 2.$$

Q5. A random sample of 10 boys had the following IQ's:

70, 120, 110, 101, 88, 83, 95, 98, 107, 100. Do these data support the assumption of population mean IQ of 100 at 5% level of significance? (Tabulated value = 2.26)

Q6. In a hurdle race, a player has to cross 10 hurdles. The probability that he will clear each hurdle is $\frac{5}{6}$. What is the probability that he will knock down fewer than 2 hurdles?

Q7. If $f(z) = u + iv$ is an analytic function and $u + v = \frac{x}{(x^2 + y^2)}$, find $f(z)$.

Part -C

[Marks: 12 each]

Q8.(a) Evaluate $\int_0^{2\pi} \frac{\cos 3\theta}{(5 - 4\cos \theta)} d\theta$ using contour integration.

OR

(b) Evaluate $\int_C \frac{12z - 7}{(z - 1)^2(2z + 3)} dz$, where C is $|z + i| = \sqrt{3}$, using Cauchy Residue theorem.

Q9.(a) Fit a Poisson distribution to the following data and test the goodness of fit :

x	0	1	2	3	4	5
f	6	13	13	8	4	3

(Tabulated value = 9.49)

OR

(b) A random variable X has the following probability distribution :

X	-2	-1	0	1	2	3
P(X)	0.1	k	0.2	2k	0.3	k

- (i) Find k .
- (ii) Find $P(-1 < X \leq 2)$.
- (iii) Mean of X .
- (iv) Variance of X .
